

04.03 Applying Roots and Powers

Concept summary and key formulas

Key ideas

Use root and power properties to rewrite expressions into simpler equivalent forms.

Even roots require nonnegative radicands in the real-number setting.

Check whether a power is applied to the entire product or quotient.

Key formulas

$$(ab)^n = a^n b^n$$

$$\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$$

$$(a^m)^n = a^{mn}$$